Reanalysis of Volker (2022)

Network embeddedness, trust, and cooperation

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Overview

- Reproduces Table 2 of the paper.
- Source: Volker, T. B. (2022). *The future is made today: Concerns for reputation foster trust and cooperation* [Unpublished master's thesis]. Utrecht University. PDF
- 16 study-specific tests from 8 studies, each on a regression coefficient β:
  - $H_1 : \beta > 0$
  - $H_u : \beta \in \mathbb{R}$
  - $H_c : \beta \leq 0$
- Complexity $c_1^{(t)} = 0.5$ in all tests: the prior $\mathrm{N}\left(0, \hat{\Sigma}_\beta / b\right)$ is centered at 0 (Volker, 2022, pp. 32, 36).

```
library(dplyr)
library(here)

# Function that enumerates all components of the complete joint complement
source(here("functions/compute_joint_complements.r"))
```

```r
# Helpers to format numbers in the text
fmt2 <- function(x) sprintf("%.2f", x)  # 2 decimals
fmt4 <- function(x) sprintf("%.4f", x)  # 4 decimals
fmtE <- function(x) {                   # scientific notation for LaTeX math
    if (!is.finite(x)) return("\\infty")
    e <- floor(log10(abs(x)))
    sprintf("%.2f \\times 10^{%d}", x / 10^e, e)
}
```

Data

- Layout from Volker's Table 2; Bayes factors from his Table 4 (3 decimals).
- Trustfulness: proportion of the endowment sent to the trustee.
- Trustworthiness: proportion of the amount sent that is returned (p. 27).

```r
table2_df <- data.frame(
    study = c(
        rep("bolton_2004", 2),
        rep("duffy_2013_no_netmin", 2),
        rep("duffy_2013_no_netfull", 2),
        rep("buskens_2010", 2),
        rep("van_miltenburg_2012", 2),
        rep("frey_2019", 2),
        rep("barrera_buskens_2009", 2),
        "seinen_schram_2006",
        "corten_2016"
    ),
    outcome = c(
        rep(c("Trustfulness", "Trustworthiness"), 7),
        "Cooperation (helping)",
        "Cooperation"
    ),
    no_network = c(
        0.67, 0.69, 0.62, 0.68, 0.75, 0.62, 0.83, 0.87,
        0.90, 0.90, 0.77, 0.67, 0.73, 0.46, 0.38, 0.49
    ),
    network = c(
        0.75, 0.61, 0.64, 0.77, 0.86, 0.94, 0.94, 0.94,
        0.88, 0.91, 0.81, 0.88, 0.70, 0.50, 0.61, 0.37
    ),
    bf_iu = c(
        1.474, 0.642, 1.292, 1.823, 1.799, 2.000, 1.848,
        1.680, 0.675, 1.115, 1.426, 1.998, 0.482, 1.450,
        1.902, 0.102
    ),
    bf_ic = c(
        2.799, 0.473, 1.825, 10.319, 8.958, 2.60e+04,
        12.168, 5.255, 0.509, 1.260, 2.486, 1.01e+03,
        0.318, 2.634, 19.351, 0.054
    )
)
```

- This is a check: in the special case where $c_1^{(t)} = 0.5$ for all studies, the result should equal the one from equal weighting.
- Volker reports $\text{BF}_{1,u}^{(t)}$ and $\text{BF}_{1,c}^{(t)}$.
- `compute_joint_complements()` also needs $\text{BF}_{c,u}^{(t)}$, which can be obtained by transitivity: $\text{BF}_{c,u}^{(t)} = \text{BF}_{1,u}^{(t)} / \text{BF}_{1,c}^{(t)}$.

```
results_df <- table2_df %>%
    transmute(
        Study   = study,
        Outcome = outcome,
        bf_1u   = bf_iu,
        bf_1c   = bf_ic,
        bf_cu   = bf_iu / bf_ic
    )

# Input for compute_joint_complements():
# P_theta_in_Hc is the complexity of the complement, 1 - c_1 = 0.5
table2_ready <- results_df %>%
    select(Study, bf_1u, bf_cu) %>%
    mutate(P_theta_in_Hc = 0.5)

T <- nrow(results_df)  # number of study-specific tests (16)
```

Study	Outcome	$\text{BF}_{1,u}$	$\text{BF}_{1,c}$	$\text{BF}_{c,u}$
bolton_2004	Trustfulness	1.474	2.799	0.527
bolton_2004	Trustworthiness	0.642	0.473	1.357
duffy_2013_no_netmif	Trustfulness	1.292	1.825	0.708
duffy_2013_no_netmif	Trustworthiness	1.823	10.319	0.177
duffy_2013_no_netfull	Trustfulness	1.799	8.958	0.201
duffy_2013_no_netfull	Trustworthiness	2.000	26000.000	0.000
buskens_2010	Trustfulness	1.848	12.168	0.152
buskens_2010	Trustworthiness	1.680	5.255	0.320
van_miltenburg_2012	Trustfulness	0.675	0.509	1.326
van_miltenburg_2012	Trustworthiness	1.115	1.260	0.885
frey_2019	Trustfulness	1.426	2.486	0.574
frey_2019	Trustworthiness	1.998	1010.000	0.002
barrera_buskens_2009	Trustfulness	0.482	0.318	1.516
barrera_buskens_2009	Trustworthiness	1.450	2.634	0.550
seinen_schram_2006	Cooperation (helping)	1.902	19.351	0.098
corten_2016	Cooperation	0.102	0.054	1.889

Product Bayes factors

```
bf_1u <- prod(results_df$bf_1u)  # against the joint unconstrained hypothesis
bf_1c <- prod(results_df$bf_1c)  # against the incomplete joint complement
```

- $\text{BF}_{1,u}^{(\text{joint})} = 7.24$ (Volker: 7.27; difference due to rounding).
- $\text{BF}_{1,c}^{(\text{joint})} = 5.23 \times 10^{11}$.

BF against the complete joint complement

- Complete joint complement: $H_1^{(t)}$ is false in at least one test.
- Computed in three ways; with exact inputs, all three agree here (with $c_1^{(t)} = 0.5$, equal and complexity weighting coincide).

1: Equal weighting, enumeration (Eq. 7)

- Sums over all $2^T - 1 = 65,535$ components with equal weights, giving $\mathrm{BF}^{(\mathrm{joint})}_{c^*,u}$.
- Then $\mathrm{BF}^{(\mathrm{joint})}_{1,c^*} = \mathrm{BF}^{(\mathrm{joint})}_{1,u} / \mathrm{BF}^{(\mathrm{joint})}_{c^*,u}$.

```
res <- compute_joint_complements(table2_ready,
                                 replace_0_with = 0.001,
                                 round_res = FALSE)

# Sanity check: BF_1u(joint) / BF_cu(joint) should equal the product of the BF_1c's
all.equal(res["BES_c", "bf_1u"] / res["BES_c", "bf_cu"], bf_1c)
```

```
[1] TRUE
```

```
bf_1c_star_enumeration <- res["BES_c", "bf_1u"] / res["BES_c_star", "bf_cu_star"]
```

2: Complexity weighting (Eq. 10)

- Check: in the special case where $c_1^{(t)} = 0.5$ for all studies, the result should equal the one obtained from equal weighting.
- Uses the $\mathrm{BF}^{(t)}_{1,u}$ and complexities $c_1^{(t)}$.

$$\mathrm{BF}^{(\mathrm{joint})}_{1,c^*} = \frac{\prod_{t=1}^{T} \mathrm{BF}^{(t)}_{1,u} \left(1 - \prod_{t=1}^{T} c_1^{(t)}\right)}{1 - \prod_{t=1}^{T} c_1^{(t)} \, \mathrm{BF}^{(t)}_{1,u}}$$

```
c1 <- 1 - table2_ready$P_theta_in_Hc  # complexity c_1^(t) of H_1 in each test

bf_1c_star_formula_complexity <-
    prod(results_df$bf_1u) * (1 - prod(c1)) / (1 - prod(c1 * results_df$bf_1u))
```

3: Equal weighting, simplified expression (Eq. 11)

- Uses only the $\mathrm{BF}^{(t)}_{1,c}$.

$$\mathrm{BF}^{(\mathrm{joint})}_{1,c^*} = \frac{2^T - 1}{\prod_{t=1}^{T} (1 + 1/\mathrm{BF}^{(t)}_{1,c}) - 1}$$

```r
bf_1c_star_formula_equal <- (2^T - 1) / (prod(1 + 1 / results_df$bf_1c) - 1)
```

Comparison

Method	Computed from	$\mathrm{BF}_{1,c^*}^{(\mathrm{joint})}$
Equal weighting, enumeration (Eq. 7)	$\mathrm{BF}_{1,u}^{(t)}$	7.2397
Complexity weighting (Eq. 10)	$\mathrm{BF}_{1,u}^{(t)}$ and $c_1^{(t)}$	7.2396
Equal weighting, simplified expression (Eq. 11)	$\mathrm{BF}_{1,c}^{(t)}$	7.2795

- Enumeration and Eq. 10: 7.24; Eq. 11: 7.28.
- The difference comes from *which* rounded Bayes factors each method uses:
  - Enumeration: numerator $\prod_t \mathrm{BF}_{1,u}^{(t)}$; denominator $\mathrm{BF}_{c^*,u}^{(\mathrm{joint})} \approx 1$.
  - Eq. 10: $\mathrm{BF}_{1,u}^{(t)}$.
  - Eq. 11: $\mathrm{BF}_{1,c}^{(t)}$.

Rounding check

- **Goal:** How much of the gap between $\mathrm{BF}_{1,u}^{(\mathrm{joint})} = 7.24$ and $\mathrm{BF}_{1,c^*}^{(\mathrm{joint})} = 7.28$ (Eq. 11) is due to rounding?
- The two are computed from different columns ($\mathrm{BF}_{1,u}$ vs. $\mathrm{BF}_{1,c}$), which Volker rounded separately to 3 decimals.
- **Logic:** With $c_1 = 0.5$, each $\mathrm{BF}_{1,u}^{(t)}$ determines its $\mathrm{BF}_{1,c}^{(t)}$ exactly, and vice versa:

$$\mathrm{BF}_{1,c}^{(t)} = \frac{\mathrm{BF}_{1,u}^{(t)}}{2 - \mathrm{BF}_{1,u}^{(t)}} \qquad \Longleftrightarrow \qquad \mathrm{BF}_{1,u}^{(t)} = \frac{2\,\mathrm{BF}_{1,c}^{(t)}}{1 + \mathrm{BF}_{1,c}^{(t)}}$$

  - Set 1: reported $\mathrm{BF}_{1,u}^{(t)} \rightarrow$ implied $\mathrm{BF}_{1,c}^{(t)}$.
  - Set 2: reported $\mathrm{BF}_{1,c}^{(t)} \rightarrow$ implied $\mathrm{BF}_{1,u}^{(t)}$.
  - Within a set, both joint BFs use the same information, so any difference is genuine (not due to rounding).
- **Expectation**
- With $c_1^{(t)} = 0.5$, Eqs. 10 and 11 reduce to $\mathrm{BF}_{1,c^*}^{(\mathrm{joint})} = \mathrm{BF}_{1,u}^{(\mathrm{joint})} \times \text{factor}$, where $\text{factor} = (2^T - 1)/(2^T - \mathrm{BF}_{1,u}^{(\mathrm{joint})})$.

```r
# Step 1: ratio BF_1c*(joint) / BF_1u(joint) implied by Eqs. 10 and 11 when c_1 = 0.5
factor <- (2^T - 1) / (2^T - bf_1u)

# Step 2: expected BF_1c*(joint) without rounding
bf_1cstar_exp <- bf_1u * factor

# Step 3: expected difference between BF_1c*(joint) and BF_1u(joint)
diff_exp <- bf_1cstar_exp - bf_1u
```

With $T = 16$ and $\text{BF}_{1,u}^{(\text{joint})} = 7.238915$:

1. **Factor:** how much larger $\text{BF}_{1,c^*}^{(\text{joint})}$ should be than $\text{BF}_{1,u}^{(\text{joint})}$ (here 0.0095% larger):

$$\text{factor} = \frac{65535}{65536 - 7.238915} = 1.000095209$$

2. **Expected** $\text{BF}_{1,c^*}^{(\textbf{joint})}$ without rounding:

$$7.238915 \times 1.000095209 = 7.239604$$

3. **Expected within-set difference:**

$$7.239604 - 7.238915 = 0.000689 \approx 0.0007$$

- This is the genuine difference that remains without rounding. The table below compares it with the observed differences.

```
# Set 1: consistent set based on the reported BF_1u's
bf_1c_from_1u <- results_df$bf_1u / (2 - results_df$bf_1u)

check_set1_bf_1u_joint <- prod(results_df$bf_1u)
check_set1_bf_1cstar   <- (2^T - 1) / (prod(1 + 1 / bf_1c_from_1u) - 1)

# Set 2: consistent set based on the reported BF_1c's
bf_1u_from_1c <- 2 * results_df$bf_1c / (1 + results_df$bf_1c)

check_set2_bf_1u_joint <- prod(bf_1u_from_1c)
check_set2_bf_1cstar   <- (2^T - 1) / (prod(1 + 1 / results_df$bf_1c) - 1)
```

Inputs	$\text{BF}_{1,u}^{(\text{joint})}$	$\text{BF}_{1,c^*}^{(\text{joint})}$	Difference
Set 1: reported $\text{BF}_{1,u}$ values	7.2389	7.2396	0.0007
Set 2: reported $\text{BF}_{1,c}$ values	7.2788	7.2795	0.0007
Reported values (mixed sets)	7.2389	7.2795	0.0406

Conclusion:

- Within each set: difference ≈ 0.0007, as expected.
- Reported (mixed) values: difference $= 0.0406$.
- So the gap is almost entirely due to rounding, mostly from the smallest Bayes factors (e.g., Corten et al., 2016: reported $\text{BF}_{1,u} = 0.102$, but $\text{BF}_{1,c} = 0.054$ implies $\text{BF}_{1,u} = 0.1025$).

Summary

$\text{BF}_{1,u}^{(\text{joint})}$	7.24
$\text{BF}_{1,c}^{(\text{joint})}$	5.23e+11
$\text{BF}_{1,c^*}^{(\text{joint})}$	7.28

Session info

```
sessionInfo()
```

```
R version 4.6.1 (2026-06-24)
Platform: aarch64-apple-darwin23
Running under: macOS Tahoe 26.5.2

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib;  LAPACK version 3.

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: Europe/Berlin
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices datasets  utils     methods   base

other attached packages:
[1] stringr_1.6.0          tibble_3.3.1           here_1.0.2             dplyr_1.2.1
[5] BayesFactor_0.9.12-4.8 Matrix_1.7-5           coda_0.19-4.1

loaded via a namespace (and not attached):
 [1] jsonlite_2.0.0     compiler_4.6.1     renv_1.2.4         tinytex_0.61       tidyselect_1.2.1
 [6] Rcpp_1.1.2         MatrixModels_0.5-4 parallel_4.6.1     jquerylib_0.1.4    yaml_2.3.12
[11] fastmap_1.2.0      lattice_0.22-9     R6_2.6.1           generics_0.1.4     knitr_1.52
[16] rprojroot_2.1.1    bslib_0.12.0       pillar_1.11.1      rlang_1.3.0        cachem_1.1.0
[21] stringi_1.8.9      xfun_0.61          sass_0.4.10        otel_0.2.0         cli_3.6.6
[26] withr_3.0.3        magrittr_2.0.5     digest_0.6.39      grid_4.6.1         pbapply_1.7-5
[31] mvtnorm_1.4-2      lifecycle_1.0.5    vctrs_0.7.3        evaluate_1.0.5     glue_1.8.1
[36] rmarkdown_2.32     tools_4.6.1        pkgconfig_2.0.3    htmltools_0.5.9
```